Retail OS — Validated Complete Specs (All Modules)

This is the **single source of truth** for Fred's Retail OS build. It replaces and consolidates all previous spec pages.

Every module here has been validated against two sources:

- **Source A** — The 17 Segment Playbooks (what each business type needs)
- **Source B** — The Excel Master Features file (the feature inventory)

Modules are ordered by build phase. Every module has a feature table, data model, and a copy-paste AI coding prompt.

Standard AI Prompt Header — Paste at the Start of Every Coding Session

```
You are building Posterita Retail OS — an Android-native modular
POS super app. Stack: Kotlin, Jetpack Compose, Material Design 3,
Room (SQLite), WorkManager. Architecture: offline-first. All
writes go to local Room DB first, then sync via WorkManager
background job. Every entity has outlet_id for multi-outlet
support. Roles: Owner > Manager > Cashier > Staff (enforced in
code, not just UI). Currency: Rs (Mauritian Rupee) — integers
only, no decimals displayed. Payments: Cash, Peach Payments (card
terminal), Juice MCB (QR), MAUCAS (QR), Voucher, On-Account.
Receipts: ESC/POS Bluetooth thermal print OR WhatsApp send OR SMS.
Compliance: MRA e-invoicing — every receipt needs BRN, VAT ID,
datetime, unique transaction ID. Module activation: features
toggled per outlet via modules_enabled[] config array. Min SDK:
Android 10 (API 29). Target: Android 14.
```

Module Index

Phase 1 — Retail Core (Build First)

#	Module	Source	Status
1	POS & Checkout	Excel ✅ + Playbook ✅	Ready
2	Product & Catalogue	Excel ✅ + Playbook ✅	Ready
3	Inventory & Stock Control	Excel ✅ + Playbook ✅	Ready
4	Payments & Tendering	Excel ✅ + Playbook ✅	Ready
5	Invoicing, Fiscalisation & MRA	Excel ✅ + Playbook ✅	Ready
6	Receipts & Billing	Excel stub → Playbook rewrite ✅	Ready
7	Staff & Workforce	Excel ✅ + Playbook ✅	Ready
8	Customer & Loyalty	Excel ✅ + Playbook ✅	Ready
9	Promotions & Discounts	Excel ✅ + Playbook ✅	Ready
10	Reporting & Business Intelligence	Excel ✅ + Playbook ✅	Ready
11	Offline Architecture & Sync Engine	Placeholder rewrite ✅	Ready
12	Mobile Experience & Device Support	Placeholder rewrite ✅	Ready
13	Admin, Config & Localisation	Stub + Playbook rewrite ✅	Ready
14	User Management, Permissions & Audit	Placeholder rewrite ✅	Ready
15	Customer-Facing Experiences	Placeholder rewrite ✅	Ready
16	Real-Time Fraud Monitoring	Stub + Playbook rewrite ✅	Ready
17	Integrations & API Framework	Excel ✅ + Playbook ✅	Ready
18	Supplier & Purchase Order Management	Excel ✅ + Playbook ✅	Ready

Phase 2 — Restaurant & F&B (Build Second)

#	Module	Source	Status
19	Table & Floor Management	Playbook only ✅	Ready
20	Kitchen Order Ticket (KOT) & KDS	Playbook only ✅	Ready
21	Menu, Modifiers & Scheduling	Excel + Playbook ✅	Ready
22	QSR Speed Mode	Playbook only ✅	Ready
23	Order Types & Delivery	Excel + Playbook ✅	Ready
24	Batch Production Module (Bakery)	Playbook only ✅	Ready
25	Café Modifier & Loyalty Pack	Playbook only ✅	Ready

Phase 3 — Segment Extensions (Build Third)

#	Module	Source	Status
26	Pharmacy — Prescription & Compliance	Playbook only ✅	Ready
27	Spa & Salon — Appointments & Commissions	Playbook only ✅	Ready
28	Freelancers — Quote, Invoice & Calendar	Playbook only ✅	Ready
29	Vehicle Services — Plate CRM & Job Billing	Playbook only ✅	Ready
30	Field Sales — Mobile Catalogue & Credit	Playbook only ✅	Ready
31	Contractors — Job Card & Van Stock	Playbook only ✅	Ready
32	Mobile Vans — Van Ops & Replenishment	Playbook only ✅	Ready
33	Event Sales — Setup Wizard & Recap	Playbook only ✅	Ready
34	Microentrepreneurs — Simple POS	Playbook only ✅	Ready
35	Multi-Store & Franchise Operations	Placeholder rewrite ✅	Ready

Phase 4 — Future (Do Not Build Yet)

#	Module	Notes
36	Self-Checkout Kiosks	Phase 4 only
37	Experimental Features	Future roadmap

Build Checklists

Phase 1 Checklist

- [ ] Core data model + Room SQLite schema
- [ ] SyncQueue table + WorkManager background sync engine
- [ ] Module activation system (modules_enabled[] per outlet)
- [ ] Staff PIN login + role-based permission enforcement
- [ ] POS checkout screen (product grid + cart)
- [ ] Barcode scanner (camera ML Kit + BT HID)
- [ ] Multi-payment screen (cash/card/Juice/MAUCAS/voucher)
- [ ] MRA e-invoicing API integration
- [ ] Bluetooth ESC/POS receipt printing
- [ ] WhatsApp receipt sharing
- [ ] Product catalogue with variants and modifiers
- [ ] Inventory with stock deduction on sale
- [ ] Expiry and batch tracking
- [ ] Shift open/close with cash reconciliation
- [ ] Promotions engine (auto-apply + promo codes)
- [ ] Customer CRM (phone-number based, loyalty points)
- [ ] Daily dashboard + Z-report
- [ ] Peach Payments card terminal SDK integration
- [ ] Offline-first architecture validated

Phase 2 Checklist

- [ ] Table & floor management (visual map)
- [ ] KOT printing with station routing
- [ ] Kitchen Display Screen (KDS) via local WiFi
- [ ] Menu scheduling (breakfast/lunch/dinner)